

EDUCATION

Michigan State University

Ph.D. in Computer Science, Advisor: Dr. Jiayu Zhou

East Lansing, USA

Sept 2020 - June 2025

Northeastern University

Bachelor of Computer Science

Shenyang, China

Sept 2016 - June 2020

RESEARCH INTERESTS

Trustworthy AI: Out-of-domain generalization/detection, Robustness, Federated learning, Security, LLMs, Healthcare

EXPERIENCE

Samsung Electronics America

Machine Learning Scientist

June 2025 -

Project: Develop machine learning methods for content-aware ads recommendation systems

Michigan State University

Graduate Research Assistant

August 2020 - May 2025

Advisor: Dr. Jiayu Zhou

Topic: Develop robust and trustworthy machine learning algorithms across domains

GE Healthcare

Research Intern

May 2024 - August 2024

Advisor: Dr. Danica Xiao

Project: Enhance long-tail knowledge of LLMs for question answering (QA) tasks from different domains, especially healthcare domain

JD Explore Academy

Research Intern

May 2021 - August 2021

Advisor: Dr. Fengxiang He

Project: Domain generalization for audio data from different domains

Data Science Lab at University of Pittsburgh

Student Intern in Electrical and Computer Engineering Department

July 2019 - August 2019

Advisor: Dr. Bin Gu & Dr. Heng Huang

Project: The Model Selection for Semi-supervised Support Vector Machines via Solution Path Algorithms

PUBLICATIONS

- **Adaptive Factual Grounded Information-Seeking for Active Reasoning.**
Siqi Liang, Jiankun Wang, **Shuyang Yu**, Raed Al Kontar, Jiayu Zhou. 2026.
Under review.
- **ConDS: Context Distribution Shift for Robust In-Context Learning.**
Shuyang Yu, Sumyeong Ahn, Siqi Liang, Bairu Hou, Jiabao Ji, Shiyu Chang, Jiayu Zhou. 2025.
Under review, preprint link.
- **Dynamic Uncertainty Ranking: Enhancing In-Context Learning for Long-Tail Knowledge in LLMs.**
Shuyang Yu, Runxue Bao, Parminder Bhatia, Taha Kass-Hout, Jiayu Zhou, Cao Xiao. 2025.
Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers) (NAACL). 2025. link.

- **Fast Online Adaptive Neural MPC via Meta-Learning.**
Yu Mei, Xinyu Zhou, **Shuyang Yu**, Vaibhav Srivastava, Xiaobo Tan. 2025.
The 5th Modeling, Estimation and Control Conference (MECC). 2025. [link](#).
- **Safe and Robust Watermark Injection with a Single OoD Image.**
Shuyang Yu, Junyuan Hong, Haobo Zhang, Haotao Wang, Zhangyang Wang, Jiayu Zhou. 2024.
2024 International Conference on Learning Representations (ICLR). 2024. [link](#).
- **Global Model Selection for Semi-Supervised Support Vector Machine via Solution Paths.**
Yajing Fan*, **Shuyang Yu***, Bin Gu, Ziran Xiong, Zhou Zhai, Heng Huang, Yi Chang. 2024.
IEEE Transactions on Neural Networks and Learning Systems (TNNLS), 2024. [link](#).
- **Who Leaked the Model? Tracking IP Infringers in Accountable Federated Learning.**
Shuyang Yu, Junyuan Hong, Yi Zeng, Fei Wang, Ruoxi Jia, Jiayu Zhou. 2023.
NeurIPS 2023 Workshop on Regulatable ML. preprint [link](#).
- **Revisiting Data-Free Knowledge Distillation with Poisoned Teachers.**
Junyuan Hong*, Yi Zeng*, **Shuyang Yu***, Lingjuan Lyu, Ruoxi Jia, Jiayu Zhou. 2023.
Proceedings of the 40th International Conference on Machine Learning (ICML). 2023. [link](#).
- **Turning the Curse of Heterogeneity in Federated Learning into a Blessing for Out-of-Distribution Detection.**
Shuyang Yu, Junyuan Hong, Haotao Wang, Zhangyang Wang, Jiayu Zhou. 2023.
2023 International Conference on Learning Representations (ICLR) (Spotlight, top-25% of accepted papers). 2023. [link](#).
- **Robust Unsupervised Domain Adaptation from a Corrupted Source.**
Shuyang Yu, Zhuangdi Zhu, Boyang Liu, Anil Jain, Jiayu Zhou. 2022.
2022 IEEE International Conference on Data Mining (ICDM). IEEE, 2022: 1299–1304. [link](#).
- **Federated Adversarial Debiasing for Fair and Transferable Representations.**
Junyuan Hong, Zhuangdi Zhu, **Shuyang Yu**, Zhangyang Wang, Hiroko H. Dodge, Jiayu Zhou. 2021.
Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining (KDD), 617–627. [link](#).
- **A kernel path algorithm for general parametric quadratic programming problem.**
Bin Gu, Ziran Xiong, **Shuyang Yu**, Guansheng Zheng. 2021.
Pattern Recognition, 2021, 116: 107941. [link](#).
- **Tackle Balancing Constraint for Incremental Semi-Supervised Support Vector Learning.**
Shuyang Yu*, Bin Gu*, Kumpeng Ning, Haiyan Chen, Jian Pei, Heng Huang. 2019.
Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining (KDD), 1587–1595. [link](#).

*equal contribution

SELECTED RESEARCH PROJECTS

Advertising machine learning: revenue prediction, bidding estimation, and incrementality measurement **2025 - Present**

- **Revenue prediction modeling:** I designed and improved large-scale machine learning models for advertising revenue prediction, including baseline training pipelines, model architecture optimization, and multi-task learning strategies to improve predictive performance under noisy and imbalanced data conditions.
- **Bidding price estimation:** I developed bidding price estimation models using adaptive re-weighting and adaptive quantile regression methods to handle wide-range target distributions, improve estimation accuracy, and enhance model robustness across different bidding scenarios.

- **Causal inference for incrementality measurement:** I applied causal inference methods, including incrementality measurement, to evaluate advertising effectiveness, identify key performance drivers, and analyze treatment effects across different ads, user segments, and business dimensions.

Large Language Models (LLMs):

2023 - Present

- **Adaptive information-seeking for active reasoning:** In training-free active reasoning methods, we reformulated follow-up question selection as a transition-graph search and proposed Adaptive Factual Grounded Information-Seeking, which selectively grounds simulated answers with factual descriptions to reduce noisy transitions and improve task success.
- **Robustness against noisy samples in the augmented candidate set:** I proposed a context distribution shift method to improve the quality of candidate sets used for augmenting LLMs. The method is compatible with different retrievers and enhances their robustness against noisy or irrelevant retrieved samples.
- **Enhancing the long-tail knowledge in LLMs:** I investigated the limitations of existing retrieval-augmented in-context learning methods for long-tail question answering. To address this challenge, I proposed a reinforcement learning-based dynamic uncertainty ranking method that leverages LLM feedback and a budget controller to select more informative retrieved candidates.

Trustworthiness: intellectual property (IP) protection

2022 - Present

IP protection can prevent the potential financial and security impact caused by illegal reproduction or duplication of high-value models. I explored watermarking for IP protection both in a centralized and a distributed setting.

- **Watermarking without training data:** I proposed a novel watermark method based on OoD data, which fills in the gap of backdoor-based IP protection of deep models without training (ID) data. The proposed watermark method is both sample efficient and time efficient without sacrificing the model utility.
- **Tracking IP infringers:** To make the model leakage from anonymity to accountability, I proposed a watermarking method that enables ownership verification and leakage tracing at the same time for distributed system with hundreds of clients.

Trustworthiness: defense against backdoor attacks

2022 - Present

Backdoor attacks can alter model behaviors drastically in the presence of pre-designed triggers but remains silent on clean samples.

- Uncovered the security risk of data-free knowledge distillation (KD) regarding untrusted pre-trained teacher models on diverse backdoor types distillation methods.
- Identified two potential causes for the backdoor infiltrating from the teacher to the student via data-free KD, that may inspire defense methods. To mitigate the data-free backdoor transfer, I also proposed the plug-in defensive method for data-free KD methods.

Out-of-distribution detection

2021 - Present

Given an out-of-distribution (OoD) test sample that does not belong to any training classes, deep neural networks (DNNs) may predict it as one of the training classes with high confidence, which is doomed to be wrong. OoD detection methods can distinguish OoD samples from in-distribution (ID) samples to avoid this phenomenon.

- Proposed a novel federated OoD synthesizer to take advantage of data heterogeneity to facilitate OoD detection in federated learning (FL), allowing a client to learn external class knowledge from other non-iid federated collaborators in a privacy-aware manner.
- Bridged a critical research gap since OoD detection for FL is currently not yet well-studied in literature.

Domain adaptation

2021 - Present

Unsupervised Domain Adaptation (UDA) provides a promising solution for learning without supervision, which transfers knowledge from relevant source domains to target domain.

- Proposed a learning paradigm that served as a general framework to integrate existing UDA approaches to improve their robustness and is powered with rigorous theoretical justifications.
- The proposed learning framework can be flexibly combined with available UDA approaches that are orthogonal to our work to improve their robustness under corrupted data.

TALKS & PRESENTATIONS

- (Invited talk) Enhancing the Adaptiveness and Reliability of Machine Learning Models in Diverse Domains. *Machine Learning seminar, Samsung Electronics America*, 2025.
- (Poster presentation) Dynamic Uncertainty Ranking: Enhancing In-Context Learning for Long-Tail Knowledge in LLMs. *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL 2025)*.
- (Poster presentation) Who Leaked the Model? Tracking IP Infringers in Accountable Federated Learning. *Michigan AI Symposium*, 2024.
- (Poster presentation) Safe and Robust Watermark Injection with a Single OoD Image. *2024 International Conference on Learning Representations (ICLR 2024)*.
- (Poster presentation) Who Leaked the Model? Tracking IP Infringers in Accountable Federated Learning. *NeurIPS 2023 Workshop on Regulatable ML*, 2023.
- (Poster presentation) Revisiting Data-Free Knowledge Distillation with Poisoned Teachers. *Proceedings of the 40th International Conference on Machine Learning (ICML 2023)*.
- (Spotlight presentation) Turning the Curse of Heterogeneity in Federated Learning into a Blessing for Out-of-Distribution Detection. *2023 International Conference on Learning Representations (ICLR 2023)*.
- (Short presentation) Robust Unsupervised Domain Adaptation from a Corrupted Source. 2022 IEEE International Conference on Data Mining (**ICDM 2022**).

PROFESSIONAL ACTIVITIES

Journal/Conference Review:

- Neural Information Processing Systems (NeurIPS 2025)
- International Conference on Knowledge Discovery & Data Mining (KDD 2024, KDD 2022)
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS)
- IEEE Transactions on Knowledge and Data Engineering (TKDE)
- SIAM International Conference on Data Mining (SDM)
- Neurocomputing
- ACM Transactions on Computing for Healthcare

Program Committee Member:

- European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD 2026)
- FedKDD: International Joint Workshop on Federated Learning for Data Mining and Graph Analytics (FedKDD 2024).

Conference Volunteer:

- 2022 IEEE International Conference on Data Mining (ICDM), Orlando, FL.

HONORS & AWARDS

ICDM travel award	2022
Outstanding Graduate Award	2020
First Prize in China Undergraduate Mathematical Contest in Modeling (Liaoning Provincial Division)	2018
School “Excellent Student” Award	2016-2017
Second-class Scholarship	2017-2018
First-class Scholarship	2016-2017
National Scholarship (top 1%)	2016-2017